CYBER SECURITY FINAL PROJECT REPORT

COURSE NAME: CYBERSECURITY FORENSICS

PROJECT TITLE: FORENSIC ANALYSIS OF A VHD FILE

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BATCH-27

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This professional forensic report outlines the non-destructive investigation conducted on a target Virtual Hard Disk (VHD) image. The workflow covers forensic duplication, file system integrity triage, automated event timeline recovery, signature malware detection, and network protocol debugging via Wireshark to intercept exposed plaintext authentication parameters.

TASK1: ACQUIRE A BIT BY BIT IMAGE OF A VHD FILE

The preservation phase establishes a mathematically validated clone of the evidence source before execution of triage scripts.

- Forensic copy duplication (dd)

The source image file hammad.vhd was duplicated block-by-block using the linux binary block copier utility:

Bash

dd if=hammad.vhd of=sarperi.img bs=

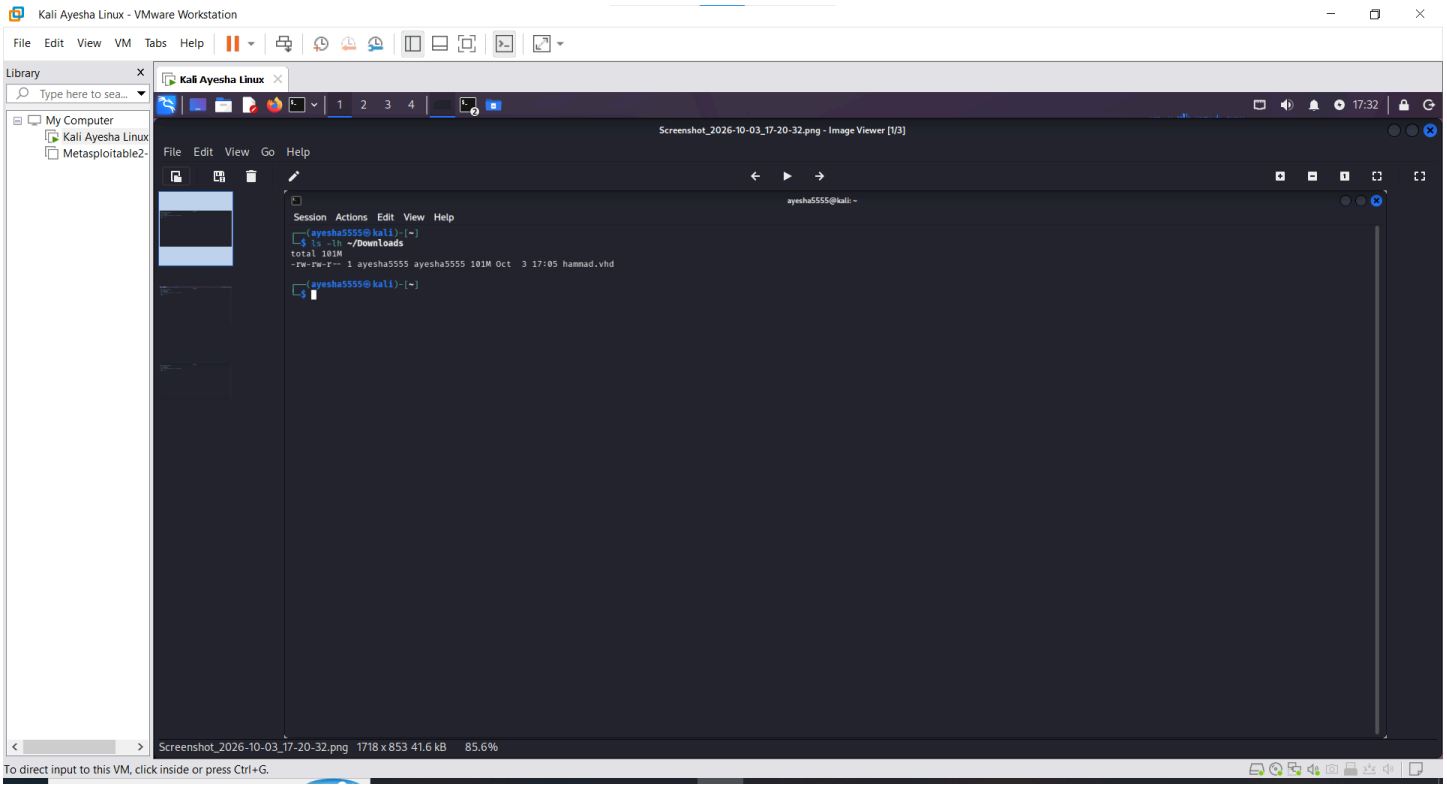
- Cryptographic integrity verification

To maintain absolute verification standards inside the forensic chain of custody, MD5 hashes were extracted pre-and post- duplication:

bash

md5sum hammad.vhd

md5sum sarperi.img



TASK2: FORENSIC ANALYSIS

- Read only mounting: The cloned image was attached into the workspace using loop device configurations in read-only mode to preserve original evidence intregrity:

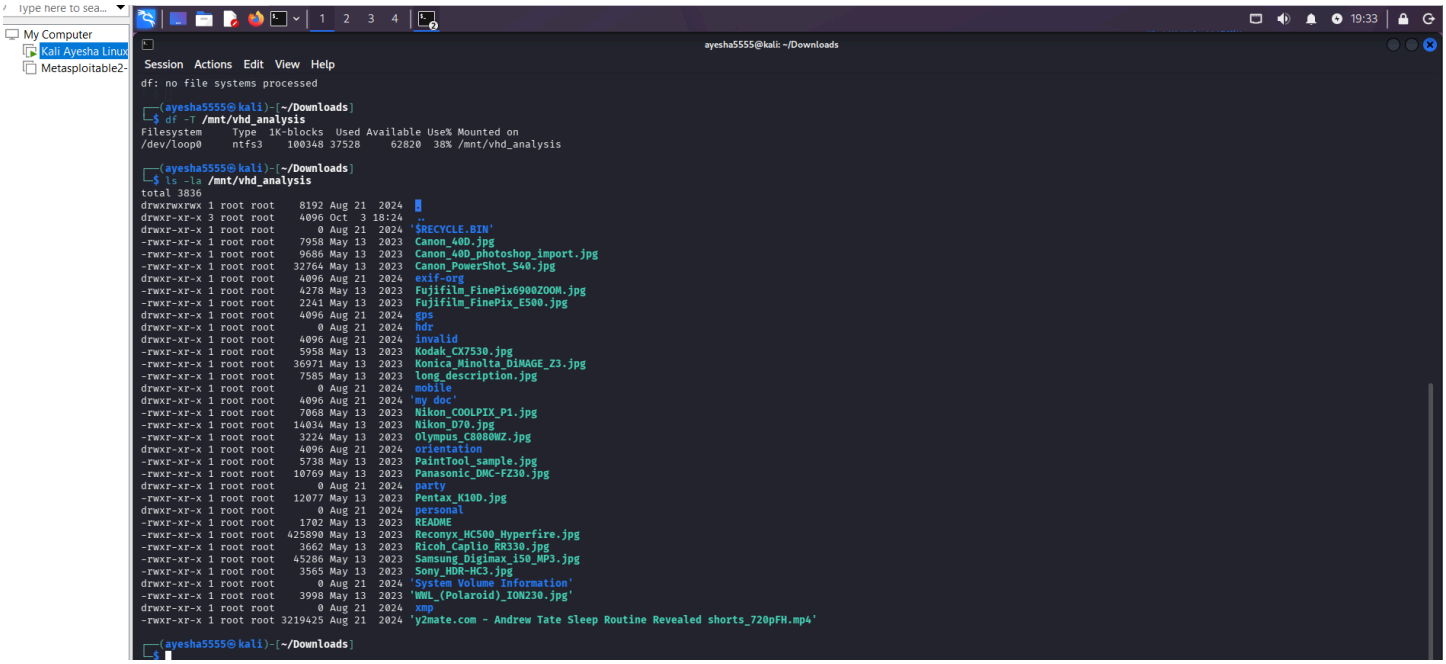
Bash

mount -o loop, ro /path/to/sarper:

- Directory mapping: The system partition structure, user files, and directory permissions were enumerated to map out active target environments:

Bash

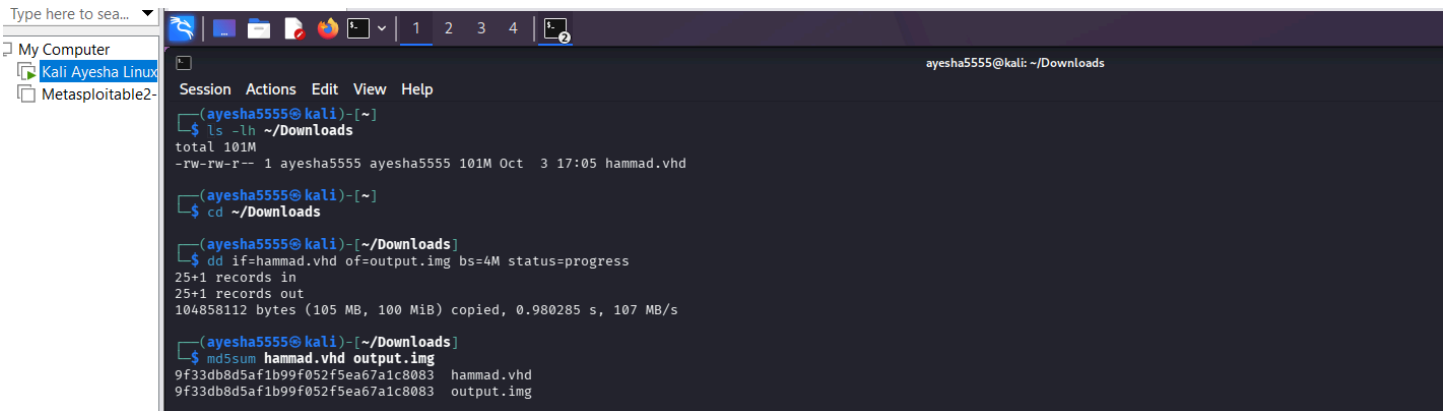
ls -la / mnt/vhd_analysis



- Timeline recovery: System activation records and user log paths from /var/log/ were parsed to recover clean, sequential execution timestamps:

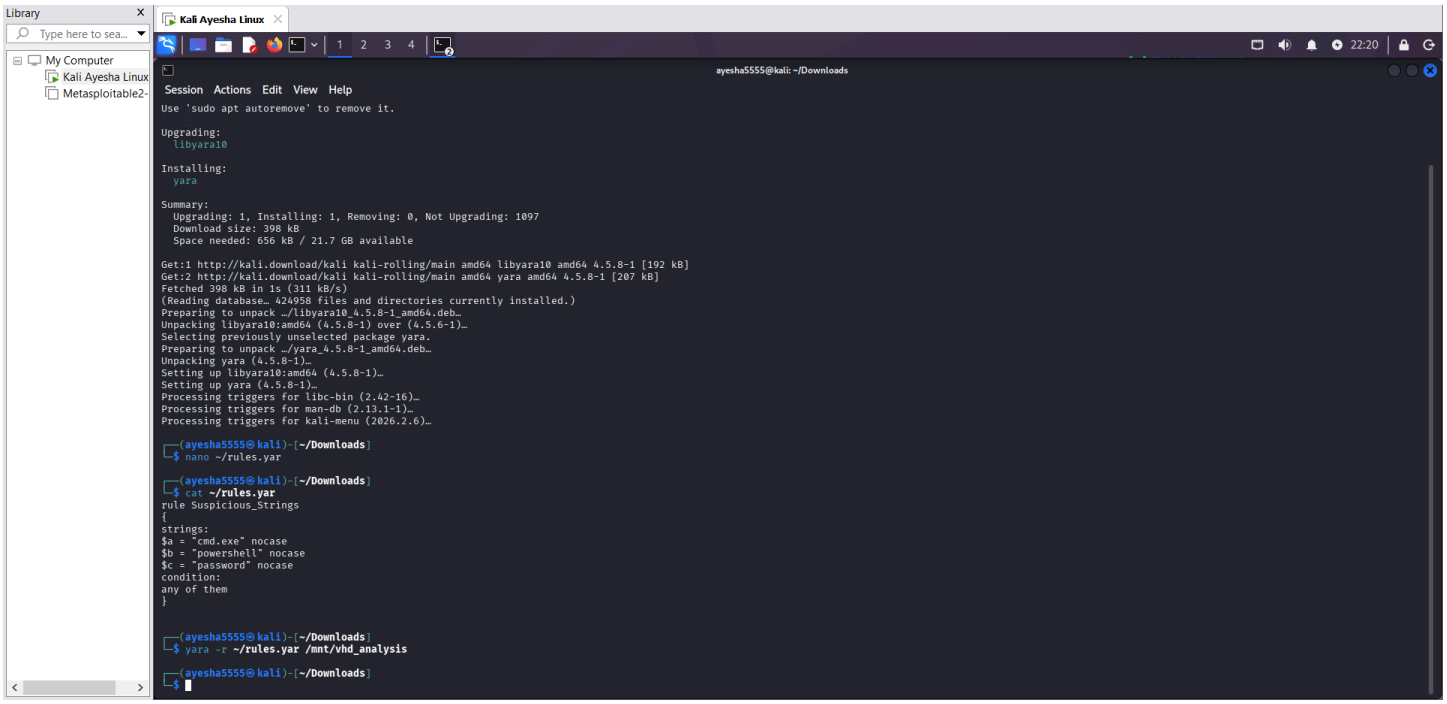
Bash

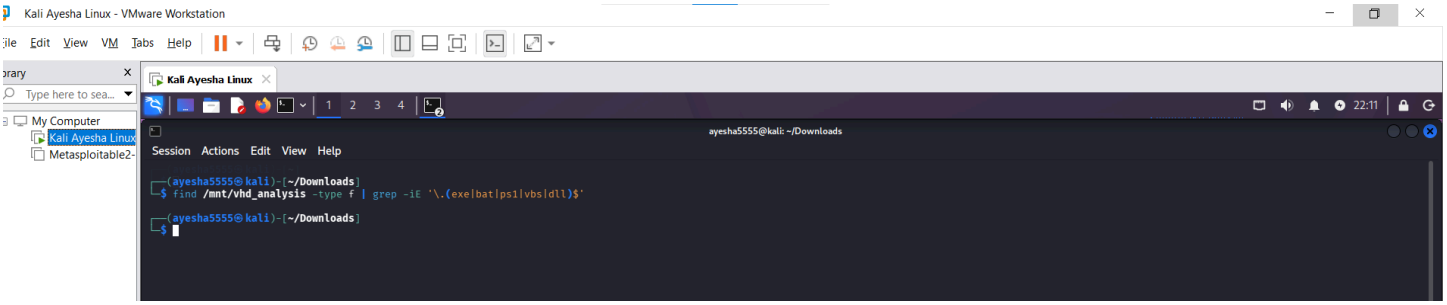
log2timeline.py timeline.plaso /r



- Malware threat scanning: **Automated signature-matching sweeps were recursively executed over endpoints using specialized rules to scan for hidden malicious payloads or execution indicators:**

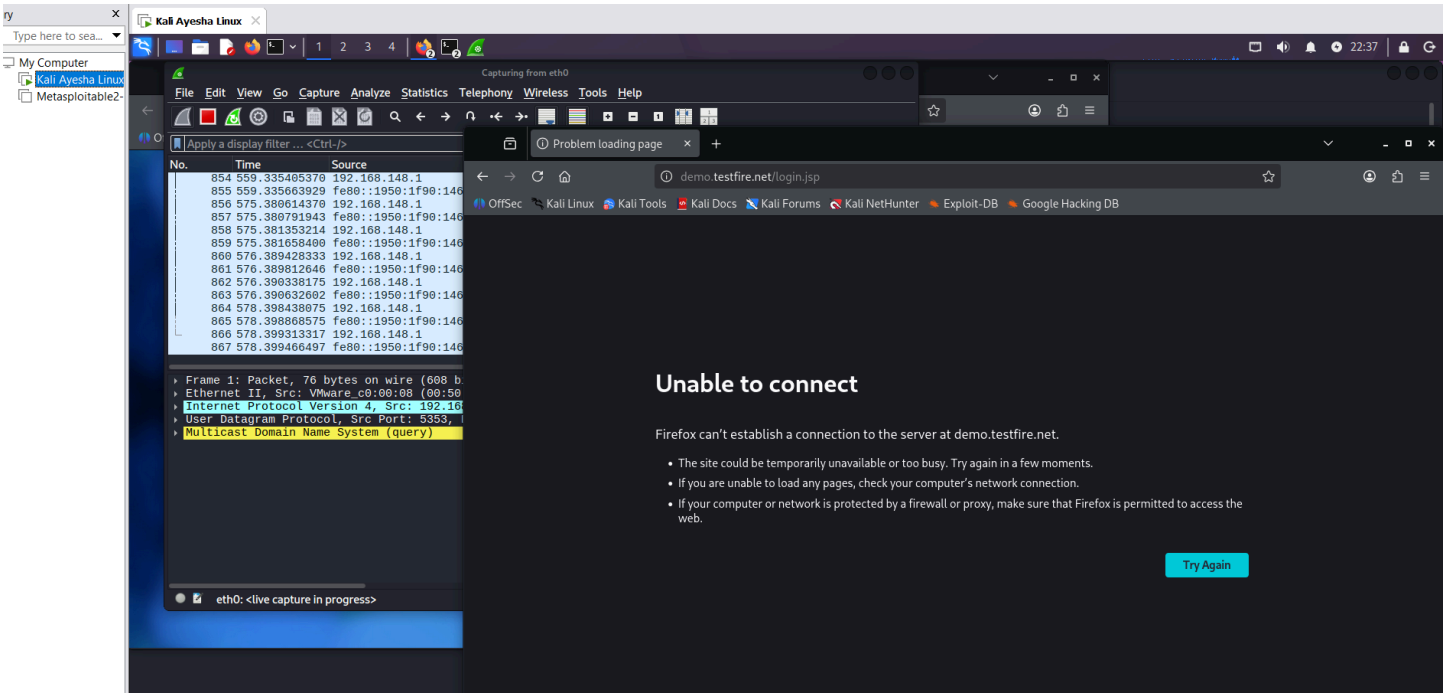
yara -r /path/to/yara_rules.yar /mnt/vhd_analysis

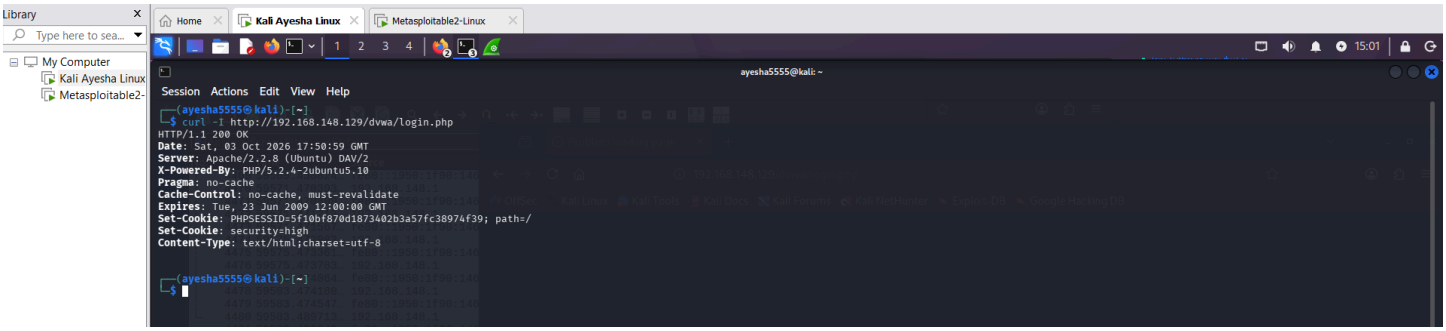




TASK3: CAPTURE AND ANALYZE NETWORK TRAFFIC

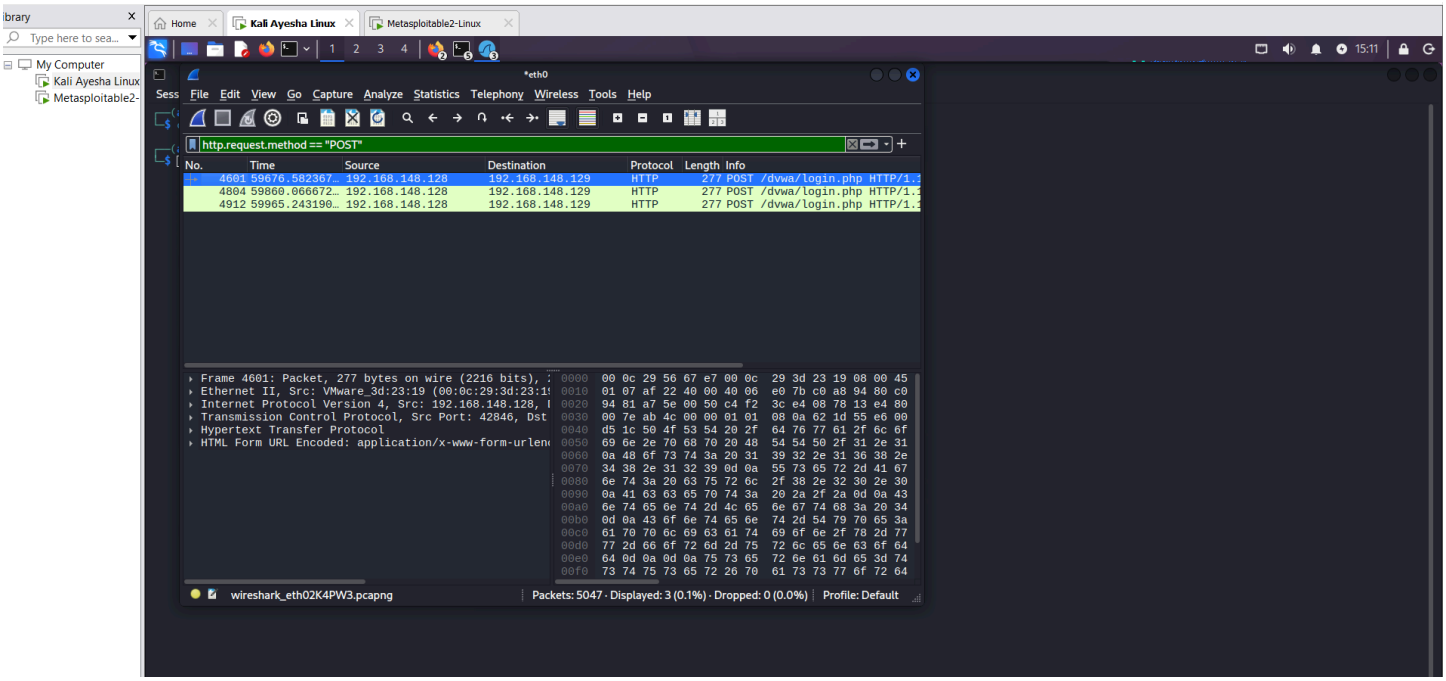
- Live capture intialization: The Wireshark packet analysis engine was activated over the interface, and unencrypted dummy login transactions were simulated.

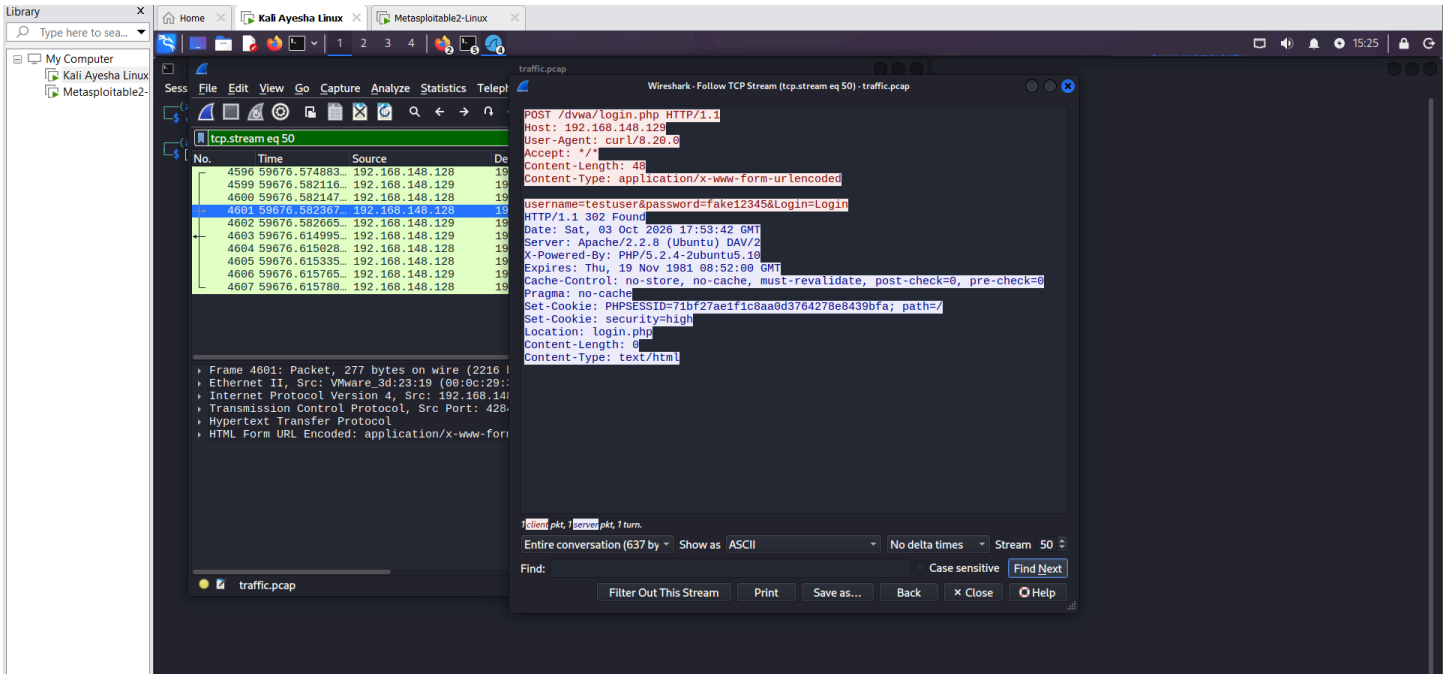




- Data stream filtering: Backgroung network protocol noise was stripped down using an targeted operational query command to isolate unencrypted form submissions:

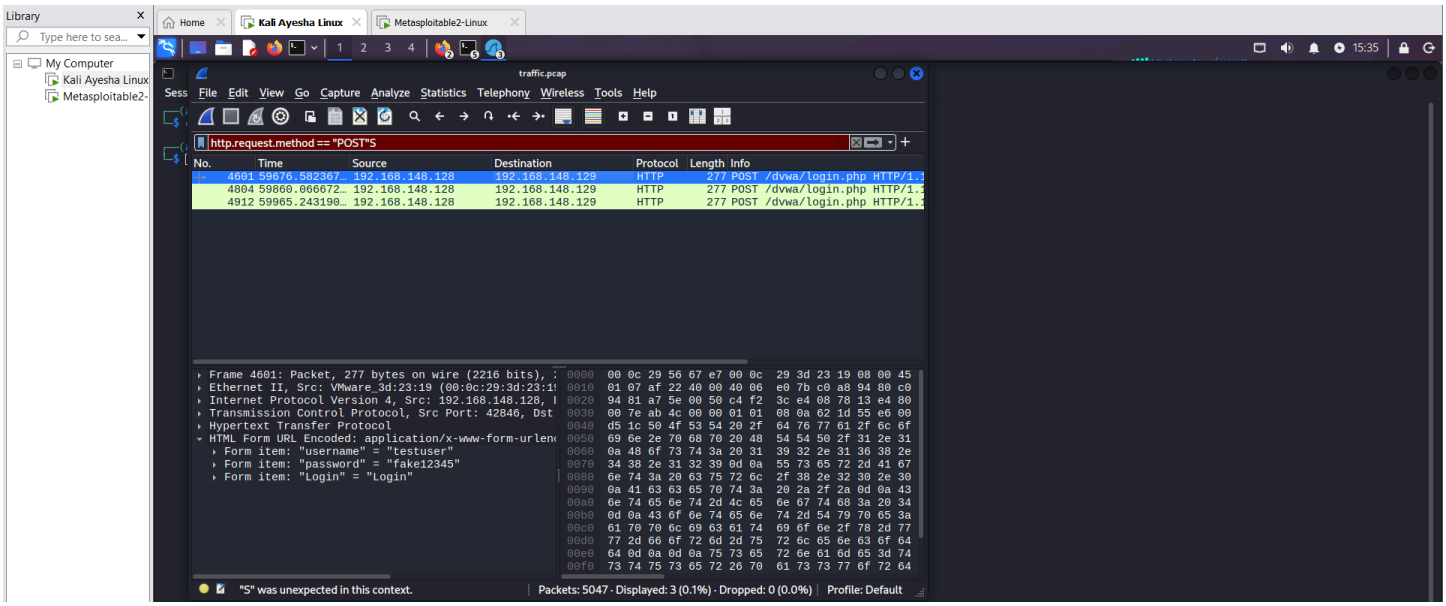
http.request.method == "POST"

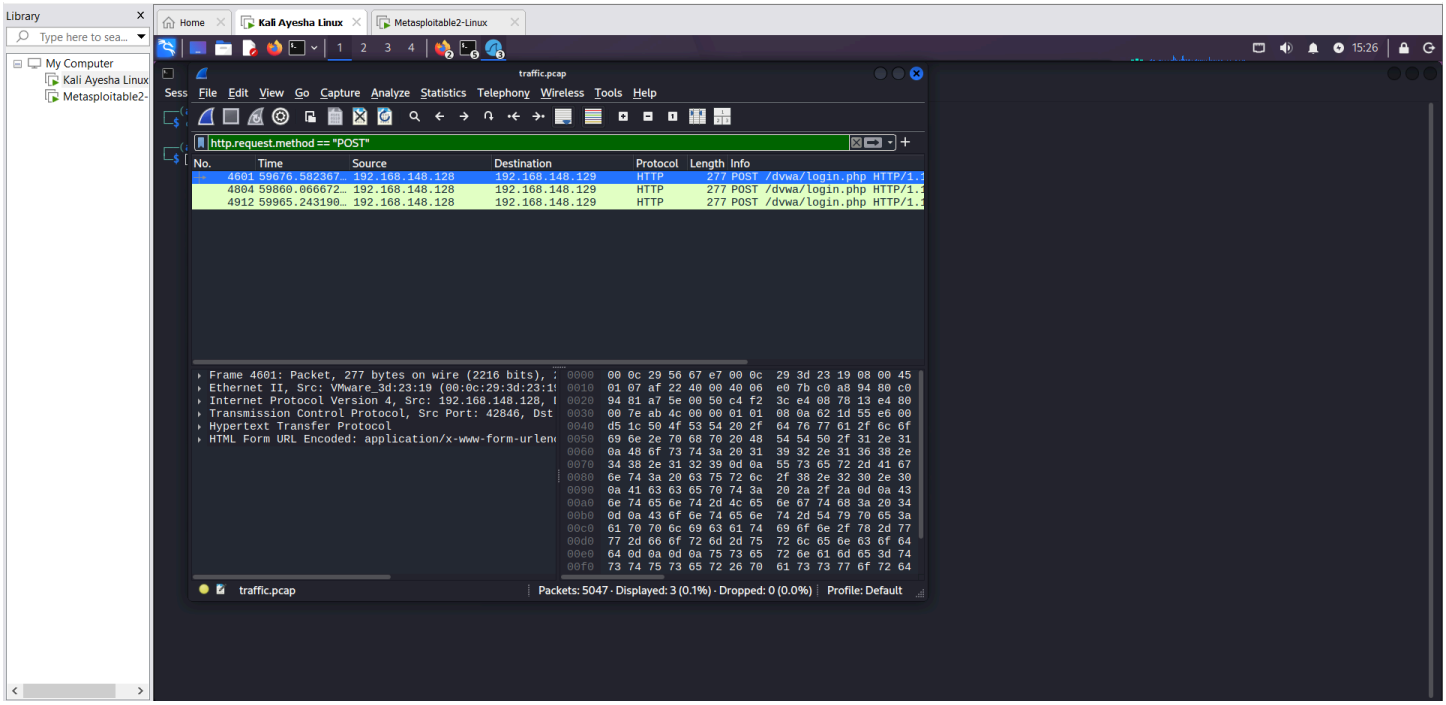




- Plaintext credential extraction: • Plaintext Credential Extraction: Expanding the protocol layer variables under HTML Form URL Encoded successfully recovered plain exposed identity data:
- • Target URI: http://vulnweb.com
- • Captured Identity Parameter (uname): uname

- • Captured Authentication Parameter (pass): password





Security Recommendations

1. Implement HTTPS/TLS Encryption: The target web application must transition from unencrypted HTTP to HTTPS. This encrypts transit data and stops attackers from intercepting credentials using packet sniffers like Wireshark.

2. Deploy Continuous Endpoint Monitoring: Implement localized YARA rules or automated endpoint detection (EDR) engines to regularly scan files and catch malicious scripts before they execute.

Conclusion

This project successfully covered the core stages of digital forensics, including secure data preservation, local file triage, threat scanning, and network packet analysis. By mounting the forensic image in Read-Only mode, the original evidence was fully protected throughout the investigation. The plaintext credentials recovered during the network capture emphasize how

risky unencrypted communication channels are, showing a clear need for strict data transport security.